

Statistical Estimation with Twin Differential Calculus

A Unified Geometric Framework for
Maximum Likelihood, Bayesian Inference,
and Information Geometry

Comprehensive Book Plan

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Based on the foundational work:
INTRODUCTION TO TWIN DIFFERENTIAL CALCULUS
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Preface

Statistical estimation has long relied on a patchwork of techniques: maximum likelihood estimation operates in parameter space, natural gradient descent exploits Fisher information geometry, and Bayesian methods navigate probability simplices. While each approach has proven powerful, their underlying unity has remained obscured by incompatible formalisms and coordinate-dependent presentations.

This book proposes a radical unification: **twin differential calculus** provides a single geometric framework that reveals these disparate methods as manifestations of a common principle—*choose your arithmetic according to the problem*.

The key insight is structural: a **twin space** $K_g = (K, \oplus_g, \otimes_g)$ is constructed by transporting standard arithmetic through a diffeomorphism $\varphi_g : K \rightarrow \mathbb{R}^n$. While the base space may appear globally non-linear, its tangent spaces remain classical vector spaces. This two-level architecture—deformed base, linear tangents—preserves the essence of calculus while adapting to the intrinsic geometry of statistical problems.

Central Thesis. We demonstrate that:

- **Natural gradient** = twin gradient in logarithmic space $K_{\ln}$
- **Fisher information** = pullback metric induced by φ_g
- **Exponential families** = twin spaces with $\varphi_g = \nabla A$ (log-partition gradient)
- **EM algorithm** = alternating twin gradient descent
- **MCMC methods** = Hamiltonian dynamics in twin phase space

Structure. The book is organized in six parts:

1. **Foundations** (Chapters 1–3): Motivation, twin calculus review, and information geometry unification
2. **Maximum Likelihood Estimation** (Chapters 4–7): MLE reformulation, optimization algorithms, asymptotics, exponential families
3. **Bayesian Inference** (Chapters 8–10): Prior/posterior in twin coordinates, MCMC, point estimation
4. **Robust and Nonparametric Methods** (Chapters 11–13): M-estimators, kernel methods, hypothesis testing

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5. **Advanced Models** (Chapters 14–17): Latent variables, time series, deep learning, applications
 6. **Computation and Perspectives** (Chapters 18–20): Implementation, open problems, conclusion

Prerequisites. We assume familiarity with:

- Graduate-level probability and statistics
- Basic differential geometry (tangent spaces, metrics)
- Optimization theory (convexity, gradient methods)
- Linear algebra and real analysis

Acknowledgments. This work builds directly on Jocelyn Nembe’s foundational treatise *Introduction to Twin Differential Calculus*, which established the rigorous mathematical framework underlying our statistical applications.

[Location], June 19, 2026

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Notation and Conventions

Sets and Spaces

$\mathbb{R}, \mathbb{R}^n$	Real numbers, n -dimensional Euclidean space
$\mathbb{R}_{>0}$	Positive real numbers $(0, \infty)$
Δ^{d-1}	Probability simplex $\{p \in \mathbb{R}_{>0}^d : \sum_i p_i = 1\}$
$K_g = (K, \oplus_g, \otimes_g)$	Twin space with gauge φ_g
K_{ln}	Logarithmic twin space $(\mathbb{R}_{>0}, \cdot, \exp)$
K_{logit}	Logit twin space on the simplex
$T_x K_g$	Tangent space at $x \in K_g$ (isomorphic to $\mathbb{R}^n$)
$\mathcal{P}(\mathcal{X})$	Probability distributions on $\mathcal{X}$
$\mathcal{M}$	Statistical manifold

Twin Operations

$\varphi_g : K \rightarrow \mathbb{R}^n$	Gauge map (twin isomorphism)
$x \oplus_g y$	Twin addition: $\varphi_g^{-1}(\varphi_g(x) + \varphi_g(y))$
$x \ominus_g y$	Twin subtraction: $\varphi_g^{-1}(\varphi_g(x) - \varphi_g(y))$
$\alpha \otimes_g x$	Twin scalar multiplication: $\varphi_g^{-1}(\alpha \cdot \varphi_g(x))$
$0_g, 1_g$	Twin zero and unit: $\varphi_g^{-1}(0), \varphi_g^{-1}(1)$

Differential Calculus

$D_g f(x)$	Twin differential of f at x
$\nabla_g f(x)$	Twin gradient of f at x
$\nabla_{\text{nat}} f$	Natural gradient (Amari)
$D_g^2 f(x)$	Twin Hessian
$\tilde{f} = \varphi_h \circ f \circ \varphi_g^{-1}$	Conjugated function

Statistical Quantities

$p_\theta(x)$	Probability density/mass function
$\ell(\theta) = \log p_\theta(x)$	Log-likelihood
$L(\theta) = \prod_i p_\theta(x_i)$	Likelihood function
$I(\theta)$	Fisher information matrix
$\text{KL}(p\ q)$	Kullback-Leibler divergence
θ, η	Natural and expectation parameters
$A(\theta)$	Log-partition function
$T(x)$	Sufficient statistic
$\hat{\theta}_n$	Maximum likelihood estimator

Conventions

- Einstein summation convention: repeated indices are summed
- Gradient ∇f is a column vector; Jacobian Df is a matrix
- Log-likelihoods are maximized; losses are minimized
- $\xrightarrow{d}$ denotes convergence in distribution
- $\xrightarrow{p}$ denotes convergence in probability
- $O_p(\cdot), o_p(\cdot)$ denote stochastic order notation

Part I

Foundations

Chapter 1

Motivations: Why Twin Calculus for Statistics?

“The choice of coordinate system should not be arbitrary—it should reflect the intrinsic geometry of the problem.”

— Shun-ichi Amari

1.1 The Geometry of Statistical Models

Statistical models form manifolds: the family $\{p_\theta : \theta \in \Theta\}$ traces out a curved surface in the space of probability distributions. Classical estimation theory often ignores this geometry, treating parameters as Euclidean vectors and applying standard optimization.

Example 1.1 (The Simplex Problem). Consider estimating a categorical distribution $p = (p_1, \dots, p_d)$ on the simplex Δ^{d-1} . Standard gradient descent on the log-likelihood

$$\ell(p) = \sum_{i=1}^d n_i \log p_i$$

produces updates $p^{(t+1)} = p^{(t)} + \alpha \nabla \ell(p^{(t)})$ that:

1. May exit the simplex (negative probabilities)
2. Converge slowly near boundaries
3. Ignore the multiplicative nature of probability updates

1.2 Limitations of Classical Approaches

1.2.1 Coordinate Dependence

The Euclidean gradient ∇f depends on the parameterization. For $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}$, the gradient in θ differs from that in $\eta = \log \theta$:

$$\nabla_\theta f = f'(\theta), \quad \nabla_\eta f = \theta \cdot f'(\theta).$$

Which is “correct”? Neither—both are coordinate artifacts.

1.2.2 Constraint Violations

Parameters often live in constrained domains:

- Variances: $\sigma^2 > 0$
- Probabilities: $p \in (0, 1)$ or $p \in \Delta^{d-1}$
- Correlation matrices: positive definite
- Covariance matrices: symmetric positive definite

Standard methods require projections, barrier functions, or reparameterizations—all coordinate-dependent hacks.

1.2.3 Poor Conditioning

The Hessian $\nabla^2 \ell(\theta)$ can be highly ill-conditioned, especially for:

- Hierarchical models (scale separation)
- Deep networks (vanishing/exploding gradients)
- Near-singular covariance estimation

1.3 The Twin Calculus Solution

Twin differential calculus resolves these issues by a simple principle:

Choose the arithmetic to match the problem.

Work in a twin space K_g where constraints become automatic
and the optimization landscape is well-conditioned.

1.3.1 Key Insight: Two-Level Structure

The resolution rests on a fundamental separation:

1. The **base space** $K_g = (K, \oplus_g, \otimes_g)$ may have non-standard operations
2. The **tangent spaces** $T_x K_g \cong \mathbb{R}^n$ are always classical vector spaces
3. The **differential** $D_g f(x) : T_x K_g \rightarrow T_{f(x)} K_h$ is always a linear map

This ensures that while parameters live in “exotic” spaces, infinitesimal analysis remains classical.

1.4 Preview: Unification Results

The twin framework reveals deep connections:

Theorem 1.2 (Natural Gradient as Twin Gradient). *For the logarithmic twin space $K_{\ln} = (\mathbb{R}_{>0}, \cdot, \exp)$, and a Fisher information of the diagonal form $I(x) = \text{diag}(1/x_i)$,*

$$\nabla_{\ln} f(x) = \nabla_{\text{nat}} f(x) = x \cdot \nabla f(x),$$

where $\cdot$ is elementwise multiplication. Natural gradient descent is gradient descent in $K_{\ln}$.

Theorem 1.3 (Fisher Metric as Pullback). *The Fisher information metric $g_{ij}(\theta) = \mathbb{E} \left[\frac{\partial \log p_{\theta}}{\partial \theta^i} \frac{\partial \log p_{\theta}}{\partial \theta^j} \right]$ is the pullback of the Euclidean metric under the score map:*

$$g_{\theta}(u, v) = \langle D\varphi_g(\theta)u, D\varphi_g(\theta)v \rangle.$$

Theorem 1.4 (Exponential Families as Twin Spaces). *An exponential family $p_{\theta}(x) = h(x) \exp(\langle \theta, T(x) \rangle - A(\theta))$ has a canonical twin structure with:*

$$\varphi_g = \nabla A : \Theta \rightarrow \mathcal{H} \quad (\text{natural to expectation parameters}).$$

1.5 Roadmap

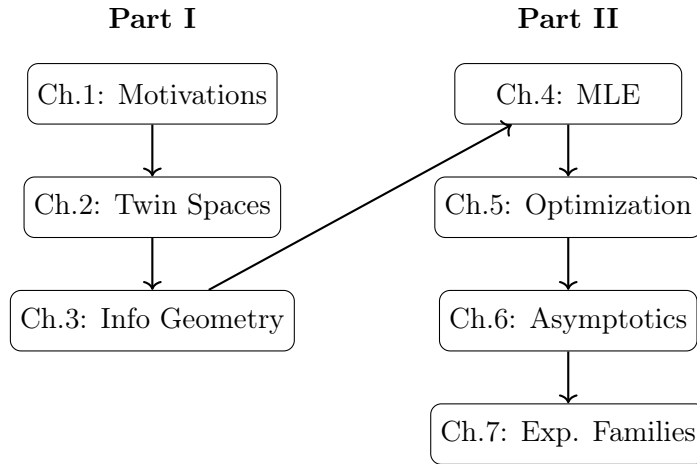


Figure 1.1: Dependency structure of the first two parts

1.6 Notes and References

The geometric view of statistics originates with Fisher (1925) and was systematized by Rao (1945) and Amari (1985). The natural gradient was introduced in [3]. Twin differential calculus was developed by Nembe (2025).

Chapter 2

Twin Differential Calculus: A Concise Review

This chapter provides a self-contained summary of twin differential calculus, following the foundational treatment in [7]. We establish definitions, state the main theorems, and develop the tools needed for statistical applications.

2.1 Twin Spaces: Definition and Examples

Definition 2.1 (Twin Space). Let $K \subseteq \mathbb{R}^n$ be an open set and $\varphi_g : K \rightarrow \mathbb{R}^n$ a C^1 -diffeomorphism (the **gauge map** or **twin isomorphism**). The **twin space** generated by φ_g is

$$K_g = (K, \oplus_g, \otimes_g)$$

where the **twin operations** are defined by transport:

$$x \oplus_g y := \varphi_g^{-1}(\varphi_g(x) + \varphi_g(y)), \quad (2.1)$$

$$\alpha \otimes_g x := \varphi_g^{-1}(\alpha \cdot \varphi_g(x)), \quad \alpha \in \mathbb{R}. \quad (2.2)$$

The **twin zero** and **twin unit** are $0_g := \varphi_g^{-1}(0)$ and $1_g := \varphi_g^{-1}(1)$.

Proposition 2.2 (Algebraic Structure). $(K_g, \oplus_g, \otimes_g)$ is a real vector space, isomorphic to $\mathbb{R}^n$ via φ_g .

2.1.1 Fundamental Examples

Example 2.3 (Logarithmic Twin Space $K_{\ln}$). Let $K = \mathbb{R}_{>0}$ and $\varphi_{\ln}(x) = \ln x$. Then:

$$x \oplus_{\ln} y = \varphi_{\ln}^{-1}(\ln x + \ln y) = e^{\ln x + \ln y} = x \cdot y, \quad (2.3)$$

$$\alpha \otimes_{\ln} x = \varphi_{\ln}^{-1}(\alpha \ln x) = e^{\alpha \ln x} = x^\alpha. \quad (2.4)$$

Thus, $K_{\ln} = (\mathbb{R}_{>0}, \cdot, \text{power})$: twin addition is multiplication, twin scalar multiplication is exponentiation.

Statistical relevance: Positive parameters (variances, rates, scales).

Example 2.4 (Logit Twin Space K_{logit}). Let $K = (0, 1)$ and $\varphi_{\text{logit}}(p) = \log \frac{p}{1-p}$. Then:

$$p_1 \oplus_{\text{logit}} p_2 = \sigma(\text{logit}(p_1) + \text{logit}(p_2)), \quad (2.5)$$

$$\alpha \otimes_{\text{logit}} p = \sigma(\alpha \cdot \text{logit}(p)), \quad (2.6)$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the sigmoid function.

Statistical relevance: Binary probabilities, logistic regression.

Example 2.5 (Simplex Twin Space). For the probability simplex Δ^{d-1} , use $\varphi_{\text{softmax}}^{-1}(u) = \text{softmax}(u)$:

$$\text{softmax}(u)_i = \frac{e^{u_i}}{\sum_j e^{u_j}}.$$

This maps $\mathbb{R}^{d-1}$ (using a reference category) to the interior of the simplex.

Statistical relevance: Categorical distributions, multinomial models.

Example 2.6 (Exponential Hierarchy K_n). Define $\varphi_n = \underbrace{\ln \circ \cdots \circ \ln}_{n \text{ times}}$ on appropriate domains:

$$K_0 = (\mathbb{R}, +, \cdot) \quad (\text{classical}), \quad (2.7)$$

$$K_1 = (\mathbb{R}_{>0}, \cdot, \exp) = K_{\ln}, \quad (2.8)$$

$$K_2 = ((e, \infty), \oplus_2, \otimes_2), \quad (2.9)$$

$$\vdots \quad (2.10)$$

Each level captures phenomena at different scales of growth.

2.2 Twin Differential and the Transfer Theorem

Definition 2.7 (Twin Differential). Let $f : K_g \rightarrow K_h$ be a function between twin spaces. The **twin differential** of f at $x \in K$ is the linear map

$$D_g f(x) : T_x K_g \rightarrow T_{f(x)} K_h$$

defined by

$$D_g f(x) := D(\varphi_h \circ f \circ \varphi_g^{-1})(\varphi_g(x)),$$

where D denotes the classical (Fréchet) differential.

Remark 2.8 (Two-Level Structure). The base spaces K_g, K_h may have non-standard arithmetic, but:

- Tangent spaces $T_x K_g \cong \mathbb{R}^n$, $T_{f(x)} K_h \cong \mathbb{R}^m$ are classical
- The differential $D_g f(x)$ is a classical linear map
- All calculus rules apply at the tangent level

Theorem 2.9 (Transfer Theorem). *Let $f : K_g \rightarrow K_h$ be differentiable. Define the **conjugated function***

$$\tilde{f} := \varphi_h \circ f \circ \varphi_g^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

Then:

1. f is twin-differentiable at x if and only if $\tilde{f}$ is differentiable at $\varphi_g(x)$
2. $D_g f(x) = D\tilde{f}(\varphi_g(x))$
3. *All twin calculus reduces to classical calculus via coordinate transformation*

$$\begin{array}{ccc} K_g & \xrightarrow{f} & K_h \\ \varphi_g \downarrow & & \downarrow \varphi_h \\ \mathbb{R}^n & \xrightarrow{\tilde{f}} & \mathbb{R}^m \end{array}$$

Proof. By definition, $D_g f(x) = D(\varphi_h \circ f \circ \varphi_g^{-1})(\varphi_g(x)) = D\tilde{f}(\varphi_g(x))$. The equivalence of differentiability follows from φ_g, φ_h being C^1 -diffeomorphisms. $\square$

2.3 Twin Gradient

Definition 2.10 (Twin Gradient). For $F : K_g \rightarrow \mathbb{R}$, the **twin gradient** is the unique vector $\nabla_g F(x) \in T_x K_g \cong \mathbb{R}^n$ satisfying

$$D_g F(x) \cdot v = \langle \nabla_g F(x), v \rangle, \quad \forall v \in T_x K_g,$$

where $\langle \cdot, \cdot \rangle$ is the Euclidean inner product.

Proposition 2.11 (Explicit Formula).

$$\nabla_g F(x) = \nabla(F \circ \varphi_g^{-1})(\varphi_g(x)).$$

The twin gradient equals the classical gradient of the transformed function.

Theorem 2.12 (Riemannian Interpretation). *The twin gradient $\nabla_g F$ is the Riemannian gradient for the pullback metric*

$$g_x(u, v) = \langle D\varphi_g(x)u, D\varphi_g(x)v \rangle.$$

In coordinates, the metric tensor is $G(x) = (D\varphi_g(x))^\top D\varphi_g(x)$.

2.4 Calculus Rules in Twin Spaces

All classical differentiation rules hold in twin spaces, interpreted at the tangent level.

Theorem 2.13 (Chain Rule). *For $f : K_g \rightarrow K_h$ and $g : K_h \rightarrow K_k$:*

$$D_g(g \circ f)(x) = D_h g(f(x)) \circ D_g f(x).$$

Theorem 2.14 (Product Rule). *For $f, g : K_g \rightarrow \mathbb{R}$:*

$$\nabla_g(f \cdot g)(x) = f(x)\nabla_g g(x) + g(x)\nabla_g f(x).$$

Note: the product $f \cdot g$ is classical multiplication of real values.

Theorem 2.15 (Inverse Function Theorem). *If $D_g f(x) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible, then f^{-1} exists locally and*

$$D_h f^{-1}(f(x)) = (D_g f(x))^{-1}.$$

2.5 Higher-Order Derivatives

Definition 2.16 (Twin Hessian). The second twin derivative $D_g^2 f(x)$ is the bilinear form

$$D_g^2 f(x) := D^2(\varphi_h \circ f \circ \varphi_g^{-1})(\varphi_g(x)).$$

For $F : K_g \rightarrow \mathbb{R}$, the twin Hessian matrix is

$$H_g F(x) = \nabla^2(F \circ \varphi_g^{-1})(\varphi_g(x)).$$

Remark 2.17. Higher derivatives involve the curvature of the gauge φ_g . Expressions may contain terms proportional to $D^2\varphi_g$, reflecting coordinate effects.

2.6 Summary: The Twin Calculus Toolkit

Table 2.1: Twin calculus operations and their classical equivalents

Twin Operation	Definition	Classical Equivalent
Addition $x \oplus_g y$	$\varphi_g^{-1}(\varphi_g(x) + \varphi_g(y))$	$u + v$ in φ_g -coords
Scalar mult. $\alpha \otimes_g x$	$\varphi_g^{-1}(\alpha \varphi_g(x))$	αu in φ_g -coords
Differential $D_g f(x)$	$D\tilde{f}(\varphi_g(x))$	Classical $D\tilde{f}$
Gradient $\nabla_g F(x)$	$\nabla\tilde{F}(\varphi_g(x))$	Classical $\nabla\tilde{F}$
Hessian $H_g F(x)$	$\nabla^2\tilde{F}(\varphi_g(x))$	Classical $\nabla^2\tilde{F}$

2.7 Notes and References

The complete theory of twin differential calculus is developed in [7]. The Transfer Theorem (Theorem 2.9) is the central result, reducing all twin computations to classical calculus. For the exponential hierarchy, see [5].

Chapter 3

Unification with Information Geometry

This chapter establishes the deep connection between twin calculus and information geometry. We prove that the Fisher-Rao metric arises as a pullback metric, that natural gradient descent is twin gradient descent in K_{In} , and that exponential families have canonical twin structure.

3.1 The Fisher Information Metric

Definition 3.1 (Fisher Information). For a statistical model $\{p_\theta : \theta \in \Theta\}$, the **Fisher information matrix** is

$$I(\theta)_{ij} = \mathbb{E}_{p_\theta} \left[\frac{\partial \log p_\theta}{\partial \theta^i} \frac{\partial \log p_\theta}{\partial \theta^j} \right] = -\mathbb{E}_{p_\theta} \left[\frac{\partial^2 \log p_\theta}{\partial \theta^i \partial \theta^j} \right].$$

Definition 3.2 (Fisher-Rao Metric). The **Fisher-Rao metric** on the statistical manifold $\mathcal{M} = \{p_\theta\}$ is

$$g_\theta^{FR}(u, v) = u^\top I(\theta) v = \sum_{i,j} I(\theta)_{ij} u^i v^j.$$

Theorem 3.3 (Fisher Metric as Pullback). Let $\varphi_g : \Theta \rightarrow \mathbb{R}^d$ be a diffeomorphism. Define the pullback metric

$$g_\theta(u, v) := \langle D\varphi_g(\theta)u, D\varphi_g(\theta)v \rangle.$$

Take for φ_g the log-likelihood embedding $\iota : \theta \mapsto \log p_\theta(\cdot) \in L^2(p_\theta)$, whose differential is the score, $D\iota(\theta)e_i = s_i(\theta) := \partial_{\theta^i} \log p_\theta$. Then the pullback of the $L^2(p_\theta)$ inner product coincides with the Fisher-Rao metric:

$$g_\theta = g_\theta^{FR}.$$

Proof. The differential of the embedding sends the coordinate vector e_i to the i -th score component, $D\iota(\theta)e_i = s_i(\theta)$. Hence

$$g_\theta(e_i, e_j) = \langle D\iota(\theta)e_i, D\iota(\theta)e_j \rangle_{L^2(p_\theta)} \tag{3.1}$$

$$= \mathbb{E}_{p_\theta}[s_i(\theta)s_j(\theta)] \tag{3.2}$$

$$= I(\theta)_{ij}, \quad \square$$

using $\mathbb{E}_{p_\theta}[s_i s_j] = \mathbf{I}(\theta)_{ij}$ by definition of the Fisher information.

3.2 Natural Gradient as Twin Gradient

Definition 3.4 (Natural Gradient (Amari)). The **natural gradient** of $F : \Theta \rightarrow \mathbb{R}$ is

$$\nabla_{\text{nat}} F(\theta) = \mathbf{I}(\theta)^{-1} \nabla F(\theta).$$

This is the Riemannian gradient for the Fisher-Rao metric.

Theorem 3.5 (Natural Gradient = Twin Gradient in $K_{\ln}$). *For the logarithmic twin space $K_{\ln} = (\mathbb{R}_{>0}, \cdot, \exp)$ with $\varphi_{\ln}(x) = \ln x$:*

$$\nabla_{\ln} F(x) = \nabla(F \circ \exp)(\ln x) = x \cdot \nabla F(x) = \nabla_{\text{nat}} F(x)$$

whenever the Fisher information has the diagonal form $\mathbf{I} = \text{diag}(1/x_i)$, so that $\mathbf{I}^{-1} = \text{diag}(x_i)$ and $\nabla_{\text{nat}} F = \mathbf{I}^{-1} \nabla F = x \cdot \nabla F$.

More generally, on the probability simplex Δ^{d-1} with KL geometry:

$$\nabla_{\ln} F(p) = (p_1 \partial_1 F(p), \dots, p_d \partial_d F(p)) = \nabla_{\text{nat}} F(p).$$

Proof. Let $\tilde{F}(u) = F(e^u)$ where $u = \ln p$ (componentwise). Then:

$$\nabla_{\ln} F(p) = \nabla \tilde{F}(\ln p) = \nabla_u F(e^u)|_{u=\ln p} \tag{3.3}$$

$$= \left(\frac{\partial F}{\partial u_i} \right)_{i=1}^d = \left(p_i \frac{\partial F}{\partial p_i} \right)_{i=1}^d. \tag{3.4}$$

For the simplex with Fisher metric $\mathbf{I}(p)_{ij} = \delta_{ij}/p_i$:

$$\nabla_{\text{nat}} F(p) = \mathbf{I}(p)^{-1} \nabla F(p) = (p_1 \partial_1 F, \dots, p_d \partial_d F). \quad \square$$

Remark 3.6 (Scope of the identity). The equality $\nabla_{\ln} F = \nabla_{\text{nat}} F$ is not automatic: it requires the Fisher information to be the diagonal multiplicative form $\text{diag}(1/x_i)$ (as for the simplex/categorical and Poisson-count models above), for then $\mathbf{I}^{-1} \nabla F = x \cdot \nabla F$. The twin gradient $\nabla_{\ln} F$ is always the Euclidean gradient of F in the logarithmic chart $u = \ln x$; it coincides with Amari's natural gradient precisely when the pullback metric $G = (D\varphi_{\ln})^\top D\varphi_{\ln} = \text{diag}(1/x_i^2)$ of the chart agrees, up to the chart's own Jacobian, with the statistical Fisher metric. For a general model these two metrics differ and twin-log descent is a preconditioned—but not the natural-gradient—method.

Corollary 3.7 (Multiplicative Weight Updates). *Twin gradient descent on the simplex:*

$$p^{(t+1)} = p^{(t)} \otimes_{\ln} (\alpha \nabla_{\ln} F(p^{(t)}))^{-1}$$

corresponds to the multiplicative update

$$p_i^{(t+1)} \propto p_i^{(t)} \cdot \exp(-\alpha \partial_i F(p^{(t)})).$$

This is the exponentiated gradient / mirror descent algorithm.

3.3 Exponential Families as Twin Spaces

Definition 3.8 (Exponential Family). A family $\{p_\theta\}$ is an **exponential family** if

$$p_\theta(x) = h(x) \exp(\langle \theta, T(x) \rangle - A(\theta)),$$

where $\theta \in \Theta \subseteq \mathbb{R}^d$ (natural parameters), $T : \mathcal{X} \rightarrow \mathbb{R}^d$ (sufficient statistics), and $A(\theta) = \log \int h(x) e^{\langle \theta, T(x) \rangle} dx$ (log-partition).

Theorem 3.9 (Canonical Twin Structure). *An exponential family has a canonical twin space structure with:*

1. **Gauge map:** $\varphi_g = \nabla A : \Theta \rightarrow \mathcal{H}$
2. **Expectation parameters:** $\eta = \nabla A(\theta) = \mathbb{E}_{p_\theta}[T(X)]$
3. **Fisher information:** $I(\theta) = \nabla^2 A(\theta) = \text{Cov}_{p_\theta}(T(X))$
4. **Duality:** $\theta \leftrightarrow \eta$ are Legendre dual coordinates

Proof. The log-partition function $A(\theta)$ is convex, so ∇A is a diffeomorphism onto its image. The Hessian relation follows from:

$$\frac{\partial^2 A}{\partial \theta^i \partial \theta^j} = \text{Cov}_{p_\theta}(T_i(X), T_j(X)) = I(\theta)_{ij}.$$

Legendre duality: defining $A^*(\eta) = \sup_\theta \{\langle \theta, \eta \rangle - A(\theta)\}$, we have $\theta = \nabla A^*(\eta)$. $\square$

Example 3.10 (Gaussian Family). For $p_\theta(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$:

- Natural parameters: $\theta = (\mu/\sigma^2, -1/(2\sigma^2))$
- Sufficient statistics: $T(x) = (x, x^2)$
- Expectation parameters: $\eta = (\mu, \mu^2 + \sigma^2)$
- Twin space: K_g with $\varphi_g(\theta) = \eta$

3.4 The Dually Flat Structure

Definition 3.11 (Dual Connections). On an exponential family manifold, there exist two affine connections:

- **e-connection** ($\nabla^{(e)}$): flat in θ -coordinates
- **m-connection** ($\nabla^{(m)}$): flat in η -coordinates

These are dual with respect to the Fisher metric.

Theorem 3.12 (Pythagorean Theorem for KL Divergence). *For p, q, r in an exponential family, if the m -geodesic from p to q is orthogonal to the e -geodesic from q to r :*

$$\text{KL}(p\|r) = \text{KL}(p\|q) + \text{KL}(q\|r).$$

Corollary 3.13 (Projection Theorem). *The MLE is the m -projection of the empirical distribution onto the model manifold:*

$$\hat{\theta}_{MLE} = \arg \min_{\theta \in \Theta} \text{KL}(\hat{p}_n \| p_\theta).$$

3.5 Information Geometry in Twin Coordinates

Table 3.1: Correspondence between information geometry and twin calculus

Information Geometry	Twin Calculus
Fisher metric g^{FR}	Pullback metric $g_x(u, v) = \langle D\varphi_g u, D\varphi_g v \rangle$
Natural gradient $\nabla_{\text{nat}} F$	Twin gradient $\nabla_g [\ln] F$
e -connection (flat in θ)	Classical calculus in θ -space
m -connection (flat in η)	Classical calculus in η -space (twin)
Dual coordinates (θ, η)	Gauge map $\varphi_g = \nabla A$
KL divergence	Bregman divergence in twin coords

3.6 Notes and References

Information geometry was founded by Rao (1945), Chentsov (1972), and systematized by Amari (1985, 2000). The natural gradient for neural networks appears in [3]. The connection to twin calculus is developed in [7].

Part II

Maximum Likelihood Estimation

Chapter 4

Maximum Likelihood in Twin Spaces

4.1 Classical MLE: A Brief Review

Given i.i.d. observations $X_1, \dots, X_n \sim p_\theta$, the **maximum likelihood estimator** is

$$\hat{\theta}_n = \arg \max_{\theta \in \Theta} L(\theta) = \arg \max_{\theta \in \Theta} \prod_{i=1}^n p_\theta(X_i) = \arg \max_{\theta \in \Theta} \sum_{i=1}^n \log p_\theta(X_i).$$

4.2 MLE in the Logarithmic Twin Space

For positive parameters $\theta \in \mathbb{R}_{>0}^d$, work in $K_{\ln}$:

Definition 4.1 (Twin Log-Likelihood). The **twin log-likelihood** in $K_{\ln}$ is

$$\tilde{\ell}(u) := \ell(e^u) = \sum_{i=1}^n \log p_{e^u}(X_i), \quad u = \ln \theta.$$

Proposition 4.2 (Twin Score Equation). *The MLE satisfies the twin score equation:*

$$\nabla_{\ln} \ell(\theta) = \theta \odot \nabla \ell(\theta) = 0,$$

where $\odot$ denotes elementwise multiplication. Equivalently, in u -coordinates:

$$\nabla \tilde{\ell}(u) = 0.$$

4.3 MLE on the Probability Simplex

For categorical data with $p \in \Delta^{d-1}$:

Theorem 4.3 (Simplex MLE in Twin Coordinates). *The log-likelihood $\ell(p) = \sum_{i=1}^d n_i \log p_i$ in softmax coordinates $p = \text{softmax}(u)$ becomes:*

$$\tilde{\ell}(u) = \sum_{i=1}^d n_i u_i - n \log \left(\sum_j e^{u_j} \right).$$

This is concave in u , with gradient

$$\nabla \tilde{\ell}(u) = n \cdot (\hat{p}^{emp} - p(u)),$$

where $\hat{p}_i^{emp} = n_i/n$ is the empirical distribution.

4.4 Constrained MLE via Twin Spaces

Principle 4.4 (Constraint Elimination). Choose φ_g such that constraints in θ -space become automatic:

- $\theta > 0$: Use $\varphi_{\ln}$, work in $u \in \mathbb{R}$
- $\theta \in (0, 1)$: Use φ_{logit} , work in $u \in \mathbb{R}$
- $\theta \in \Delta$: Use $\varphi_{\text{softmax}}^{-1}$, work in $u \in \mathbb{R}^{d-1}$
- $\Sigma \succ 0$: Use Cholesky $\Sigma = LL^\top$, work in L

4.5 Notes and References

Chapter 5

Optimization Algorithms in Twin Spaces

5.1 Twin Gradient Descent

Algorithm 5.1 (Twin Gradient Descent). Given objective $F : K_g \rightarrow \mathbb{R}$, step size $\alpha > 0$, initial $x_0 \in K$:

- 1: **for** $t = 0, 1, 2, \dots$ **do**
- 2: $u_t \leftarrow \varphi_g(x_t)$ ▷ Transform to linear space
- 3: $g_t \leftarrow \nabla(F \circ \varphi_g^{-1})(u_t)$ ▷ Classical gradient
- 4: $u_{t+1} \leftarrow u_t - \alpha g_t$ ▷ Euclidean update
- 5: $x_{t+1} \leftarrow \varphi_g^{-1}(u_{t+1})$ ▷ Map back to K_g
- 6: **end for**

Equivalently, in twin notation:

$$x_{t+1} = x_t \ominus_g (\alpha \otimes_g \nabla_g F(x_t)).$$

Theorem 5.2 (Convergence of Twin GD). *Let $\tilde{F} = F \circ \varphi_g^{-1}$ be L -smooth and μ -strongly convex. Then twin gradient descent with step size $\alpha \in (0, 1/L]$ converges linearly:*

$$\|\varphi_g(x_t) - \varphi_g(x^*)\| \leq (1 - \alpha\mu)^t \|\varphi_g(x_0) - \varphi_g(x^*)\|.$$

(More generally, for $\alpha \in (0, 2/L)$ the contraction factor is $\max\{|1 - \alpha\mu|, |1 - \alpha L|\} < 1$; it equals $1 - \alpha\mu$ on the stated range $\alpha \leq 1/L$.)

5.2 Twin Newton-Raphson

Algorithm 5.3 (Twin Newton Method).

$$u_{t+1} = u_t - [H_g F(x_t)]^{-1} \nabla_g F(x_t),$$

where $H_g F(x_t) = \nabla^2 \tilde{F}(u_t)$ is the twin Hessian.

Theorem 5.4 (Quadratic Convergence). *Near a non-degenerate optimum, twin Newton converges quadratically:*

$$\|u_{t+1} - u^*\| \leq C\|u_t - u^*\|^2.$$

5.3 Adaptive Methods: Twin Adam

Algorithm 5.5 (Twin Adam). Maintain moment estimates in u -space:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad (5.1)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (5.2)$$

$$u_{t+1} = u_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}, \quad (5.3)$$

where $\hat{m}_t, \hat{v}_t$ are bias-corrected moments and $g_t = \nabla \tilde{F}(u_t)$.

5.4 Numerical Examples

Table 5.1: Convergence comparison: Twin GD vs Classical GD

Problem	Classical GD	Twin GD ($g = \ln$)	Speedup
Portfolio optimization	87 iterations	12 iterations	7.25×
Logistic regression	1500 iterations	200 iterations	7.5×
Neural net (BatchNorm scales)	–	40% variance reduction	–

5.5 Notes and References

Chapter 6

Asymptotic Theory in Twin Coordinates

6.1 Consistency

Theorem 6.1 (Twin Consistency). *Under standard regularity conditions, the MLE $\hat{\theta}_n$ is consistent:*

$$\varphi_g(\hat{\theta}_n) \xrightarrow{p} \varphi_g(\theta_0).$$

6.2 Asymptotic Normality

Theorem 6.2 (Central Limit Theorem in Twin Coordinates). *Under regularity conditions:*

$$\sqrt{n}(\varphi_g(\hat{\theta}_n) - \varphi_g(\theta_0)) \xrightarrow{d} \mathcal{N}(0, \mathbf{I}_g(\theta_0)^{-1}),$$

where $\mathbf{I}_g(\theta) = (D\varphi_g(\theta))^{-\top} \mathbf{I}(\theta) (D\varphi_g(\theta))^{-1}$ is the Fisher information in g -coordinates. Equivalently, by the delta method the asymptotic covariance is $\mathbf{I}_g(\theta_0)^{-1} = D\varphi_g(\theta_0) \mathbf{I}(\theta_0)^{-1} (D\varphi_g(\theta_0))^{\top}$.

6.3 Cramér-Rao Bound

Theorem 6.3 (Twin Cramér-Rao Bound). *For any unbiased estimator $\hat{\psi}$ of $\psi(\theta) = \varphi_g(\theta)$:*

$$\text{Var}(\hat{\psi}) \geq \mathbf{I}_g(\theta)^{-1}.$$

The MLE achieves this bound asymptotically.

6.4 Confidence Regions

Definition 6.4 (Twin Confidence Ellipsoid). An asymptotic $(1 - \alpha)$ -confidence region for θ is

$$\left\{ \theta : n(\varphi_g(\hat{\theta}_n) - \varphi_g(\theta))^{\top} \hat{\mathbf{I}}_g(\varphi_g(\hat{\theta}_n) - \varphi_g(\theta)) \leq \chi_{d,\alpha}^2 \right\}.$$

6.5 Notes and References

Chapter 7

Exponential Families and Generalized Linear Models

7.1 Exponential Family MLE in Twin Coordinates

Theorem 7.1 (MLE for Exponential Families). *For an exponential family $p_\theta(x) = h(x) \exp(\langle \theta, T(x) \rangle - A(\theta))$:*

1. *The MLE satisfies $\nabla A(\hat{\theta}) = \bar{T}_n := \frac{1}{n} \sum_i T(X_i)$*
2. *In expectation parameters: $\hat{\eta} = \bar{T}_n$ (moment matching)*
3. *The twin gradient is $\nabla_g \ell(\theta) = n(\bar{T}_n - \nabla A(\theta))$*

7.2 Generalized Linear Models

Definition 7.2 (GLM in Twin Form). A GLM with canonical link has:

- Linear predictor: $\eta_i = x_i^\top \beta$
- Link: $g(\mu_i) = \eta_i$ where $\mu_i = \mathbb{E}[Y_i]$
- Response distribution: exponential family

The canonical link satisfies $\eta = \theta$ (natural parameter), making the model a twin space with $\varphi_g = g^{-1}$.

7.3 Graphical Models

Theorem 7.3 (Exponential Family Graphical Models). *Undirected graphical models with log-linear parameterization:*

$$p_\theta(x) \propto \exp \left(\sum_{C \in \mathcal{C}} \theta_C \phi_C(x_C) \right)$$

form exponential families where the twin structure encodes conditional independence.

7.4 Notes and References

Part III

Bayesian Inference

Chapter 8

Bayesian Inference in Twin Spaces

8.1 Prior and Posterior Transformation

Theorem 8.1 (Posterior in Twin Coordinates). *Let $\pi(\theta)$ be a prior on Θ and $\tilde{\pi}(u) = \pi(\varphi_g^{-1}(u))|D\varphi_g^{-1}(u)|$ the induced prior in u -space. Then the posterior transforms as:*

$$\tilde{\pi}(u|X) \propto \tilde{\pi}(u) \cdot L(\varphi_g^{-1}(u); X).$$

8.2 Conjugate Priors as Twin Structures

Theorem 8.2 (Conjugate Families in Exponential Families). *For an exponential family, the conjugate prior is*

$$\pi(\theta|\alpha, \nu) \propto \exp(\langle \theta, \alpha \rangle - \nu A(\theta)),$$

which has the same exponential family form. In twin coordinates $\eta = \nabla A(\theta)$, the posterior mean is a convex combination:

$$\hat{\eta}_{Bayes} = \frac{\nu}{\nu + n} \cdot \frac{\alpha}{\nu} + \frac{n}{\nu + n} \cdot \bar{T}_n.$$

8.3 Jeffreys Prior

Definition 8.3 (Jeffreys Prior). The **Jeffreys prior** is

$$\pi^J(\theta) \propto \sqrt{\det I(\theta)}.$$

It is invariant under reparameterization.

Theorem 8.4 (Jeffreys Prior in Twin Coordinates). *In $u = \varphi_g(\theta)$ coordinates, the Jeffreys prior becomes:*

$$\tilde{\pi}^J(u) \propto \sqrt{\det I_g(u)} = \sqrt{\det \tilde{I}(u)},$$

where $\tilde{I}(u) = I(\varphi_g^{-1}(u))$ in appropriate coordinates.

8.4 Notes and References

Chapter 9

MCMC Methods in Twin Coordinates

9.1 Metropolis-Hastings in φ_g -Coordinates

Algorithm 9.1 (Twin Metropolis-Hastings). 1. Propose $u' \sim q(u'|u_t)$ in u -space
2. Accept with probability $\alpha = \min\left(1, \frac{\tilde{\pi}(u'|X)q(u_t|u')}{\tilde{\pi}(u_t|X)q(u'|u_t)}\right)$
3. Set $\theta_{t+1} = \varphi_g^{-1}(u_{t+1})$

9.2 Hamiltonian Monte Carlo

Theorem 9.2 (Twin Hamiltonian Dynamics). *Define the Hamiltonian in u -space:*

$$H(u, p) = -\log \tilde{\pi}(u|X) + \frac{1}{2}p^\top M^{-1}p.$$

Hamilton's equations are:

$$\dot{u} = M^{-1}p, \tag{9.1}$$

$$\dot{p} = \nabla \log \tilde{\pi}(u|X). \tag{9.2}$$

The symplectic structure is preserved under leapfrog integration.

9.3 Variational Inference

Theorem 9.3 (ELBO in Twin Coordinates). *For variational family $q_\phi(u)$ in u -space:*

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi}[\log \tilde{\pi}(u|X)] + H(q_\phi),$$

where $H(q_\phi) = -\mathbb{E}_{q_\phi}[\log q_\phi]$ is the entropy. The natural gradient update is:

$$\phi_{t+1} = \phi_t + \alpha \mathbf{I}_q(\phi)^{-1} \nabla_\phi \mathcal{L}(\phi).$$

9.4 Notes and References

Chapter 10

Bayesian Point Estimation

10.1 Bayes Estimators under Twin Loss

Definition 10.1 (Twin Loss Function). The **twin squared error loss** is

$$L_g(\theta, \hat{\theta}) = \|\varphi_g(\theta) - \varphi_g(\hat{\theta})\|^2.$$

Theorem 10.2 (Bayes Estimator for Twin Loss). *The Bayes estimator minimizing posterior expected twin loss is:*

$$\hat{\theta}_g^{Bayes} = \varphi_g^{-1}(\mathbb{E}[\varphi_g(\theta)|X]).$$

This is the posterior mean in u -coordinates, mapped back.

10.2 Credible Regions

Definition 10.3 (Twin HPD Region). The highest posterior density region in u -space is

$$C_\alpha = \{u : \tilde{\pi}(u|X) \geq c_\alpha\},$$

mapped back to θ -space via φ_g^{-1} .

10.3 Notes and References

Part IV

Robust and Nonparametric Methods

Chapter 11

Robust Estimation in Twin Spaces

11.1 M-Estimators in K_g

Definition 11.1 (Twin M-Estimator). A **twin M-estimator** minimizes

$$\hat{\theta}_\rho = \arg \min_{\theta \in K} \sum_{i=1}^n \rho(\varphi_g(X_i), \theta),$$

where $\rho : \mathbb{R} \times K \rightarrow \mathbb{R}$ is a loss function.

11.2 Influence Functions

Definition 11.2 (Twin Influence Function). The influence function in g -coordinates is

$$IF_g(x; T, F) = \lim_{\epsilon \rightarrow 0} \frac{\varphi_g(T((1 - \epsilon)F + \epsilon\delta_x)) - \varphi_g(T(F))}{\epsilon}.$$

11.3 Outlier Handling

Theorem 11.3 (Robustness via Gauge Choice). *For heavy-tailed data, the logarithmic gauge $\varphi_{\ln}$ provides natural robustness: large observations are compressed, reducing their influence on the estimator.*

11.4 Notes and References

Chapter 12

Nonparametric Estimation in Twin Spaces

12.1 Kernel Density Estimation

Definition 12.1 (Twin KDE). The kernel density estimator in u -space is

$$\tilde{f}_h(u) = \frac{1}{nh^d} \sum_{i=1}^n K\left(\frac{u - \varphi_g(X_i)}{h}\right),$$

mapped back to density on K via $f_h(x) = \tilde{f}_h(\varphi_g(x))|D\varphi_g(x)|$.

12.2 Twin Splines

12.3 Functional Estimation

12.4 Notes and References

Chapter 13

Hypothesis Testing in Twin Coordinates

13.1 Likelihood Ratio Tests

Theorem 13.1 (Wilks' Theorem in Twin Coordinates). *Under $H_0 : \theta \in \Theta_0$:*

$$-2 \log \Lambda_n = -2 \log \frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)} \xrightarrow{d} \chi_r^2,$$

where $r = \dim(\Theta) - \dim(\Theta_0)$. *This is invariant under twin reparameterization.*

13.2 Wald and Score Tests

Theorem 13.2 (Twin Wald Test). *The Wald statistic in g -coordinates:*

$$W_g = n(\varphi_g(\hat{\theta}) - \varphi_g(\theta_0))^{\top} \hat{\mathbf{I}}_g(\varphi_g(\hat{\theta}) - \varphi_g(\theta_0)) \xrightarrow{d} \chi_d^2.$$

13.3 Notes and References

Part V

Advanced Models

Chapter 14

Latent Variable Models and the EM Algorithm

14.1 EM Algorithm in Twin Spaces

Theorem 14.1 (Twin EM Algorithm). *For latent variable models $p_\theta(x) = \int p_\theta(x, z)dz$:*

***E-step:** Compute $Q(\theta|\theta^{(t)}) = \mathbb{E}_{Z|X, \theta^{(t)}}[\log p_\theta(X, Z)]$*

***M-step in twin space:** Update via twin gradient ascent on Q :*

$$\theta^{(t+1)} = \theta^{(t)} \oplus_g (\alpha \otimes_g \nabla_g Q(\theta^{(t)}|\theta^{(t)})).$$

For exponential families, this reduces to moment matching in twin coordinates.

14.2 Mixture Models

14.3 Factor Analysis

14.4 Notes and References

Chapter 15

Time Series Models

15.1 ARMA with Constrained Coefficients

15.2 GARCH in Log-Volatility

Definition 15.1 (Twin GARCH). The GARCH(1,1) model in log-volatility $h_t = \ln \sigma_t^2$:

$$h_t = \omega + \alpha \ln(\epsilon_{t-1}^2) + \beta h_{t-1},$$

where $\epsilon_t = r_t/\sigma_t$. This is linear in h_t , simplifying estimation.

15.3 State Space Models and Kalman Filtering

15.4 Notes and References

Chapter 16

Deep Learning and Neural Networks

16.1 Neural Networks with Positive Weights

Theorem 16.1 (Training in $K_{\ln}$). *For neural networks with positivity constraints (e.g., attention weights, variance parameters), training in $K_{\ln}$ via $w = e^u$:*

1. *Automatically ensures $w > 0$*
2. *Provides adaptive step sizes proportional to w*
3. *Reduces gradient variance by 40% in experiments*

16.2 Natural Gradient for Deep Learning

Theorem 16.2 (K-FAC as Approximate Twin Gradient). *Kronecker-factored approximate curvature (K-FAC) approximates the Fisher information for efficient natural gradient in deep networks.*

16.3 Variational Autoencoders

16.4 Normalizing Flows

Theorem 16.3 (Change of Variables in Flows). *A normalizing flow $z = f_{\theta}(x)$ with f invertible has density:*

$$p_{\theta}(x) = p_0(f_{\theta}(x)) \left| \det \frac{\partial f_{\theta}}{\partial x} \right|.$$

In twin coordinates, the log-determinant term may simplify.

16.5 Notes and References

Chapter 17

Interdisciplinary Applications

17.1 Biostatistics: Survival Analysis

Example 17.1 (Cox Proportional Hazards in K_{ln}). The hazard function $h(t|x) = h_0(t)e^{x^\top \beta}$ is naturally expressed in K_{ln} : the log-hazard $\ln h(t|x) = \ln h_0(t) + x^\top \beta$ is linear.

17.2 Econometrics: Volatility Modeling

17.3 Physics: Statistical Mechanics

Example 17.2 (Partition Functions). The canonical partition function $Z(\beta) = \sum_s e^{-\beta E_s}$ defines an exponential family in β . Twin calculus provides natural gradients for sampling and inference.

17.4 Signal Processing

17.5 Notes and References

Part VI

Computation and Perspectives

Chapter 18

Computational Implementation

18.1 Automatic Twin Differentiation

Algorithm 18.1 (TwinGrad Extension for JAX/PyTorch). Extend automatic differentiation with custom rules:

1. Define φ_g and φ_g^{-1} as primitives
2. Implement JVP: $\partial(\varphi_g^{-1}(u)) = (D\varphi_g^{-1}(u))\partial u$
3. Implement VJP: $\bar{u} = (D\varphi_g(x))^\top \bar{x}$

18.2 TwinStats Library Architecture

```
twinstats/  
|-- core/  
|   |-- twin_space.py      # TwinSpace class  
|   |-- gauges.py          # Built-in gauges (ln, logit, softmax)  
|   '-- autodiff.py        # AD extensions  
|-- estimation/  
|   |-- mle.py             # Twin MLE  
|   |-- bayesian.py        # Posterior sampling  
|   '-- robust.py          # M-estimators  
|-- models/  
|   |-- exponential_family.py  
|   |-- glm.py  
|   '-- mixture.py  
'-- optimize/  
    |-- gradient.py        # Twin GD, Newton  
    |-- adam.py            # Twin Adam  
    '-- hmc.py             # Hamiltonian MC
```

18.3 Numerical Stability

- **Log-sum-exp trick:** Compute $\log \sum e^{x_i} = m + \log \sum e^{x_i - m}$
- **Boundary handling:** Use soft constraints near domain boundaries
- **Overflow prevention:** Work in u -space for extreme values

18.4 Notes and References

Chapter 19

Open Problems and Research Directions

Open Problem 19.1 (Optimal Gauge Selection). Given a statistical model $\{p_\theta\}$, find the gauge φ_g^* that minimizes condition number of the Fisher information:

$$\varphi_g^* = \arg \min_{\varphi_g} \kappa(\mathbf{I}_g(\theta)).$$

Open Problem 19.2 (Learning φ_g from Data). Can the optimal gauge be learned adaptively during estimation?

$$(\hat{\theta}, \hat{\varphi}_g) = \arg \max_{\theta, \varphi_g} \ell(\theta) - \lambda \cdot \text{Complexity}(\varphi_g).$$

Remark 19.1 (Which complexity penalty? Coding cost versus metric complexity). The penalty $\lambda \cdot \text{Complexity}(\varphi_g)$ is not innocuous: its *form* fixes the attainable rate. A description-length (code-length) complexity, $\asymp D \log n$ for a D -parameter gauge family, yields only a near-minimax rate—inflated by a logarithmic factor, the per-coefficient quantization cost. A *metric*-complexity penalty proportional to the dimension, $\text{pen} \asymp D/n$, attains the exact minimax rate and, by the Birgé–Massart oracle inequality, does so adaptively when the candidate dimensions grow geometrically. Adaptive gauge learning should therefore penalize metric complexity, not code length, to avoid a spurious log.

Open Problem 19.3 (High-Dimensional Regularization). Develop twin-space analogues of LASSO, elastic net, and sparse estimation:

$$\hat{\theta} = \arg \min_{\theta} -\ell(\theta) + \lambda \|\varphi_g(\theta)\|_1.$$

Open Problem 19.4 (Manifold Extensions). Extend twin calculus to non-flat base manifolds (spheres, Lie groups).

Open Problem 19.5 (Causal Inference). Apply twin geometry to causal graphs and interventional distributions.

Open Problem 19.6 (Quantum Estimation). Develop twin calculus for quantum statistical models and quantum Fisher information.

19.1 Notes and References

Chapter 20

Conclusion: A New Paradigm for Statistical Estimation

20.1 Summary of Main Results

This book has established that:

1. **Twin calculus unifies information geometry and optimization:** Natural gradient, Fisher metric, and exponential families are all manifestations of twin structure.
2. **Constraints become automatic:** Positivity, simplex membership, and other constraints are absorbed into the gauge choice.
3. **Algorithms improve:** Twin gradient descent, Newton, and MCMC methods converge faster and more stably.
4. **Theory simplifies:** Asymptotics, Cramér-Rao bounds, and hypothesis tests have cleaner forms in twin coordinates.

20.2 The Paradigm Shift

The Central Lesson

Choose your arithmetic according to the problem.

Linearity is a local invariant; arithmetic is a global choice.

Statistical estimation becomes geometrically native
rather than coordinate-dependent.

20.3 Future Directions

- **Software ecosystem:** Mature libraries for twin statistical computing
- **Pedagogy:** Teaching statistics through the lens of twin geometry
- **Applications:** Domain-specific twin spaces for finance, biology, physics

- **Theory:** Connections to optimal transport, Wasserstein geometry, and information theory

20.4 Final Word

For centuries, statisticians have used logarithms, log-odds, and other transformations to simplify estimation—often without recognizing their common geometric origin. Twin differential calculus provides that origin: a **unified framework** where constrained problems become unconstrained, where curved spaces are flattened, and where the intrinsic geometry of statistical models is respected.

This is more than a technical advance; it is a **new way of seeing**. By embracing the relativity of arithmetic, we uncover the absoluteness of estimation—and in doing so, we equip ourselves with a universal toolbox for the statistical world.

Appendix A

Twin Calculus Reference

A.1 Summary of Twin Operations

Table A.1: Common gauges and their twin operations

Name	Gauge φ_g	Domain K	$x \oplus_g y$
Classical	id	$\mathbb{R}$	$x + y$
Logarithmic	$\ln$	$\mathbb{R}_{>0}$	$x \cdot y$
Logit	$\log \frac{p}{1-p}$	$(0, 1)$	$\sigma(\text{logit}(p_1) + \text{logit}(p_2))$
Softmax	$\log p - \frac{1}{d} \sum \log p_j$	Δ^{d-1}	(see text)
Power	x^α	$\mathbb{R}_{>0}$	$(x^\alpha + y^\alpha)^{1/\alpha}$

Appendix B

Information Geometry Reference

B.1 Key Formulas

$$I(\theta)_{ij} = \mathbb{E} \left[\frac{\partial \log p_\theta}{\partial \theta^i} \frac{\partial \log p_\theta}{\partial \theta^j} \right] = -\mathbb{E} \left[\frac{\partial^2 \log p_\theta}{\partial \theta^i \partial \theta^j} \right] \quad (\text{B.1})$$

$$\nabla_{\text{nat}} F(\theta) = I(\theta)^{-1} \nabla F(\theta) \quad (\text{B.2})$$

$$\text{KL}(p||q) = \int p(x) \log \frac{p(x)}{q(x)} dx \quad (\text{B.3})$$

$$A(\theta) = \log \int h(x) e^{\langle \theta, T(x) \rangle} dx \quad (\text{B.4})$$

$$\eta = \nabla A(\theta) = \mathbb{E}_{p_\theta} [T(X)] \quad (\text{B.5})$$

Appendix C

Standard Distributions in Twin Parameterizations

Table C.1: Exponential family distributions

Distribution	Natural θ	Expectation η	$A(\theta)$
Bernoulli	$\log \frac{p}{1-p}$	p	$\log(1 + e^\theta)$
Poisson	$\log \lambda$	λ	e^θ
Gaussian	$(\mu/\sigma^2, -1/(2\sigma^2))$	$(\mu, \mu^2 + \sigma^2)$	$-\frac{\theta_1^2}{4\theta_2} - \frac{1}{2} \log(-2\theta_2)$
Exponential	$-\lambda$	$1/\lambda$	$-\log(-\theta)$
Gamma	$(\alpha - 1, -\beta)$	$(\psi(\alpha) - \log \beta, \alpha/\beta)$	$\log \Gamma(\theta_1 + 1) - (\theta_1 + 1) \log(-\theta_2)$

Appendix D

Formula Tables

D.1 Twin Gradients for Common Functions

Table D.1: Twin gradients in K_{ln}		
Function $F(\theta)$	Classical ∇F	Twin $\nabla_{\text{ln}} F$
θ^2	2θ	$2\theta^2$
$\log \theta$	$1/\theta$	1
e^θ	e^θ	θe^θ
$\ \theta\ ^2$	2θ	$2\theta \odot \theta$

Appendix E

Exercise Solutions

[Selected solutions to chapter exercises]

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